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UNDERWRITER RECOMMENDATION MEMO

True Agentic Data Fabric – Motor Insurance Underwriting

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Policy Reference : POL-6FA1F148

Product : Motor-Comprehensive

Date & Time : 22 June 2026, 04:17 UTC

Channel : batch_demo

Generated by : CALF+CLIF Causal Underwriting Engine v2.0

APPLICANT SUMMARY

Age : 48.0 years

Annual income : Rs 600,000

Vehicle age : 14.0 years

Prior claims : 0.0

RISK PROFILE

Risk tier (CLIF) : Standard

ITE estimate : \$-573

(ITE < 0 = profitable to insurer | ITE > 0 = expected net loss)

CAUSAL FACTOR ANALYSIS

(Each factor assessed from ITE decomposition + DAG weights)

1. Prior Claims Count: Low (\$178 impact)

Causal weight : 0.31 (33.2% of total risk signal)

ITE impact : \$178 (reduces expected loss)

Assessment : Prior Claims Count – Low (value: 0.00): This factor reduces expected loss by \$178, accounting for 33.2% of the total risk signal (causal DAG weight: 0.31). Impact classification: HIGH.

Zero prior claims in the assessment period – this is the strongest positive indicator and qualifies this policy for the preferred claims-free discount. This is a PRIMARY decision driver – the underwriting outcome is materially determined by this factor.

2. Credit Score: Good (\$126 impact)

Causal weight : 0.22 (23.5% of total risk signal)

ITE impact : \$126 (increases expected loss)

Assessment : Credit Score – Good (value: 766.00): This factor increases expected loss by \$126, accounting for 23.5% of the total risk signal (causal DAG weight: 0.22). Impact classification: MODERATE. Credit score of 766.00 is in the Good band (≥ 0.60). Strong credit history is a reliable proxy for policyholder responsibility and reduced moral hazard exposure. This is a SECONDARY decision factor – it influences the risk tier and premium loading.

3. Vehicle Age: Old (\$97 impact)

Causal weight : 0.17 (18.2% of total risk signal)

ITE impact : \$97 (reduces expected loss)

Assessment : Vehicle Age – Old (value: 14.00): This factor reduces expected loss by \$97, accounting for 18.2% of the total risk signal (causal DAG weight: 0.17). Impact classification: MODERATE. Vehicle age of 14 years exceeds the 8-year threshold. Increased probability of mechanical failure, reduced crash safety ratings, and higher repair costs all contribute to this factor's \$97 adverse ITE impact. This is a SECONDARY decision factor – it influences the risk tier and premium loading.

4. Engine Cc: Medium (\$57 impact)

Causal weight : 0.10 (10.7% of total risk signal)

ITE impact : \$57 (reduces expected loss)

Assessment : Engine Cc – Medium (value: 1200.00): This factor reduces expected loss by \$57, accounting for 10.7% of the total risk signal (causal DAG weight: 0.10). Impact classification: MODERATE. Engine displacement of 1200cc – standard range. No specific loading warranted on this factor. This is a SECONDARY decision factor – it influences the risk tier and premium loading.

5. Geographic Risk: Moderate (\$40 impact)

Causal weight : 0.07 (7.5% of total risk signal)

ITE impact : \$40 (reduces expected loss)

Assessment : Geographic Risk – Moderate (value: 0.50): This factor reduces expected loss by \$40, accounting for 7.5% of the total risk signal (causal DAG weight: 0.07). Impact classification: LOW. This factor shows a positive contribution of \$40 to the Individual Treatment Effect based on the causal DAG structure. This is a MINOR contributing factor – it does not materially change the underwriting outcome.

6. Age: Experienced (\$37 impact)

Causal weight : 0.13 (7.0% of total risk signal)

ITE impact : \$37 (reduces expected loss)

Assessment : Age – Experienced (value: 48.00): This factor reduces expected loss by \$37, accounting for 7.0% of the total risk signal (causal DAG weight: 0.13). Impact classification: LOW. Driver age of 48 is in the prime experience band (25-60). Lowest expected accident frequency. This factor is neutral. This is a MINOR contributing factor – it does not materially change the underwriting outcome.

UNDERWRITING DECISION

DECISION : ACCEPTED

Premium : Rs 9,000 per annum

Confidence: 76% (MODERATE-HIGH). 3/4 primary factors align with ITE direction. One or more secondary factors show mixed signals. Decision stands – underwriter spot-check recommended.

Primary reason : ITE NEGATIVE PORTFOLIO

IRDA REASON CODES

UW-03: Vehicle age increases mechanical failure probability

TERMS AND CONDITIONS

Compulsory excess : Rs 3,000

No-Claim Bonus : Eligible after 2 claim-free years

Exclusions :

- Racing and motorsport
- Use outside declared zone
- Consequential loss

Special conditions :

- Annual vehicle inspection not required

Policy validity : 12 months from policy inception

COUNTERFACTUAL ADVICE TO APPLICANT

* Vehicle: Replacing the current vehicle (age 14 years) with one under 5 years old would eliminate the IRDA UW-03 mechanical risk loading and reduce the ITE contribution from this factor by \$97.

UNDERWRITER SIGN-OFF

System recommendation : ACCEPTED

Human review required : SPOT CHECK

Underwriter signature : _____

Date : _____

Senior review (if req) : _____

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NOTE: Generated by CALF Causal Engine. All decisions are subject to human underwriter review and IRDA guidelines. Auto-approval applies only where confidence >= 85% and no IRDA adverse codes are triggered.

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